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#include <iostream>
#include <string>
using namespace std;

// Node structure
struct Contact
{
    string name;
    string phone;
    Contact* left;
    Contact* right;
};

// Create a new contact
Contact* createContact(string name, string phone)
{
    Contact* newNode = new Contact;

    newNode->name = name;
    newNode->phone = phone;
    newNode->left = nullptr;
    newNode->right = nullptr;

    return newNode;
}

// Insert contact into BST
Contact* insertContact(Contact* root, string name, string phone)
{
    if (root == nullptr)
    {
        return createContact(name, phone);
    }

    if (name < root->name)
    {
        root->left = insertContact(root->left, name, phone);
    }
    else if (name > root->name)
    {
        root->right = insertContact(root->right, name, phone);
    }

    return root;
}

// Search contact
Contact* searchContact(Contact* root, string name)
{
    if (root == nullptr || root->name == name)
    {
        return root;
    }

    if (name < root->name)
    {
        return searchContact(root->left, name);
    }
}

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    return searchContact(root->right, name);
}

// Display contacts in sorted order
void displayContacts(Contact* root)
{
    if (root != nullptr)
    {
        displayContacts(root->left);

        cout << "Name : " << root->name << endl;
        cout << "Phone : " << root->phone << endl;
        cout << "-----" << endl;

        displayContacts(root->right);
    }
}

// Main function
int main()
{
    Contact* root = nullptr;

    int choice;
    string name, phone;

    do
    {
        cout << "\n==== Contact Search Using BST =====\n";
        cout << "1. Add Contact\n";
        cout << "2. Search Contact\n";
        cout << "3. Display Contacts\n";
        cout << "4. Exit\n";

        cout << "\nEnter your choice: ";
        cin >> choice;

        switch (choice)
        {
            case 1:
                cout << "\nEnter Contact Name: ";
                cin >> name;

                cout << "Enter Phone Number: ";
                cin >> phone;

                root = insertContact(root, name, phone);

                cout << "\nContact added successfully!\n";
                break;

            case 2:
                cout << "\nEnter Name to Search: ";
                cin >> name;

                {
                    Contact* result = searchContact(root, name);

                    if (result != nullptr)

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        {
            cout << "\nContact Found!\n";
            cout << "-----\n";
            cout << "Name : " << result->name << endl;
            cout << "Phone : " << result->phone << endl;
            cout << "-----\n";
        }
        else
        {
            cout << "\nContact not found!\n";
        }
    }
    break;

case 3:
    if (root == nullptr)
    {
        cout << "\nContact list is empty!\n";
    }
    else
    {
        cout << "\n===== Contact List =====\n";
        displayContacts(root);
    }
    break;

case 4:
    cout << "\nProgram terminated successfully.\n";
    break;

default:
    cout << "\nInvalid choice!\n";
}

} while (choice != 4);

return 0;
}

```

Output

===== Contact Search Using BST =====

1. Add Contact
2. Search Contact
3. Display Contacts
4. Exit

Enter your choice: 1

Enter Contact Name: Sahil

Enter Phone Number: 9876543210

Contact added successfully!

===== Contact Search Using BST =====

1. Add Contact
2. Search Contact
3. Display Contacts
4. Exit

Enter your choice: 1

Enter Contact Name: Mayur

Enter Phone Number: 9876543211

Contact added successfully!

===== Contact Search Using BST =====

1. Add Contact
2. Search Contact
3. Display Contacts
4. Exit

Enter your choice: 1

Enter Contact Name: Gaurav

Enter Phone Number: 9876543212

Contact added successfully!

===== Contact Search Using BST =====

1. Add Contact
2. Search Contact
3. Display Contacts
4. Exit

Enter your choice: 1

Enter Contact Name: Om

Enter Phone Number: 9876543213

Contact added successfully!

===== Contact Search Using BST =====

1. Add Contact
2. Search Contact
3. Display Contacts
4. Exit

Enter your choice: 3

===== Contact List =====

Name : Gaurav

Phone : 9876543212

Name : Mayur

Phone : 9876543211

Name : Om

Phone : 9876543213

Name : Sahil

Phone : 9876543210

===== Contact Search Using BST =====

1. Add Contact
2. Search Contact
3. Display Contacts
4. Exit

Enter your choice: 2

Enter Name to Search: Sahil

Contact Found!

Name : Sahil

Phone : 9876543210

===== Contact Search Using BST =====

1. Add Contact
2. Search Contact
3. Display Contacts
4. Exit

Enter your choice: 4

Program terminated successfully.